

Lag-Conditioned Recovery of Two Rotation Rates from a Palindromic Recurrence

Meridian Calder

SAT/H(s)H Sandbox Geometry Program

(Dated: 27 September 2026)

Abstract

A trajectory generated by two independent planar rotation rates obeys an exact fourth-order palindromic recurrence. We derive the exact cosine-root discriminant as a function of sampling lag and show that adjacent-sample recovery is intrinsically driven toward a fourth-order discriminant collapse. A fixed, answer-blind lag rule is then tested numerically on a controlled four-coordinate double-rotation signal. Under the declared acquisition and noise configuration, the same structured least-squares estimator that is poorly conditioned at short lag recovers a frequency gap of 0.10 with sub-percent median relative error at the selected stride, while substantially smaller gaps cross the observed resolution frontier. The result is a mathematical and numerical identifiability statement for this signal model, not a claim about physical dynamics.

I. MODEL AND RECURRENCE

Consider

$$\mathbf{x}(t) = (a \cos \omega t, a \sin \omega t, b \cos \nu t, b \sin \nu t). \quad (1)$$

Sampling at a lag $\tau = L\Delta t$ gives characteristic roots $e^{\pm i\omega\tau}$ and $e^{\pm i\nu\tau}$. Each coordinate therefore obeys

$$c_{n+4L} - s_1 c_{n+3L} + s_2 c_{n+2L} - s_1 c_{n+L} + c_n = 0, \quad (2)$$

where

$$s_1 = 2 [\cos(\omega\tau) + \cos(\nu\tau)], \quad (3)$$

$$s_2 = 2 + 4 \cos(\omega\tau) \cos(\nu\tau). \quad (4)$$

The two cosine modes are roots of

$$q^2 - \frac{s_1}{2}q + \frac{s_2 - 2}{4} = 0. \quad (5)$$

II. LAG-CONDITIONED DISCRIMINANT

The quadratic discriminant reduces exactly to

$$\Delta_q(\tau) = [\cos(\omega\tau) - \cos(\nu\tau)]^2 \quad (6)$$

$$= 4 \sin^2 \left[\frac{(\omega + \nu)\tau}{2} \right] \sin^2 \left[\frac{(\omega - \nu)\tau}{2} \right]. \quad (7)$$

Consequently,

$$\Delta_q(\tau) = \frac{(\omega^2 - \nu^2)^2}{4} \tau^4 + O(\tau^6) \quad (8)$$

as $\tau \rightarrow 0$. Thus exact recurrence validity does not imply stable adjacent-sample identifiability. The same expression also identifies collision lags at which either sine factor vanishes; “use the largest possible lag” is therefore not an invariant prescription.

III. CONTROLLED NUMERICAL BENCHMARK

We used $a = 1.2$, $b = 0.65$, $\omega = 1.1$, $\Delta t = 0.025$, 161 samples over $[-2, 2]$, and independent coordinate noise $\mathcal{N}(0, 10^{-8})$. At each stride, a single pair (s_1, s_2) was estimated by ordinary least squares from all four coordinates under the palindromic structure of Eq. (2). No total-least-squares claim is made.

The declared target gap set was $\Delta\omega \in \{0.10, 0.03, 0.01, 0.003\}$. Before examining recovery error, the stride was selected by

$$L_\star = \arg \max_L \min_{\Delta\omega} \Delta_q(L\Delta t), \quad (9)$$

subject to retaining at least 40 recurrence windows. This gives $L_\star = 30$.

IV. USE AND LIMITS

For the SAT/H(s)H sandbox, Eq. (7) supplies a controlled front end for reconstructing paired rotation rates before any proposed translation into frame-history or the Hagalaz transform machinery. Outside that context, the same derivation is a generic identifiability result for any sampled two-frequency signal with the palindromic characteristic polynomial above.

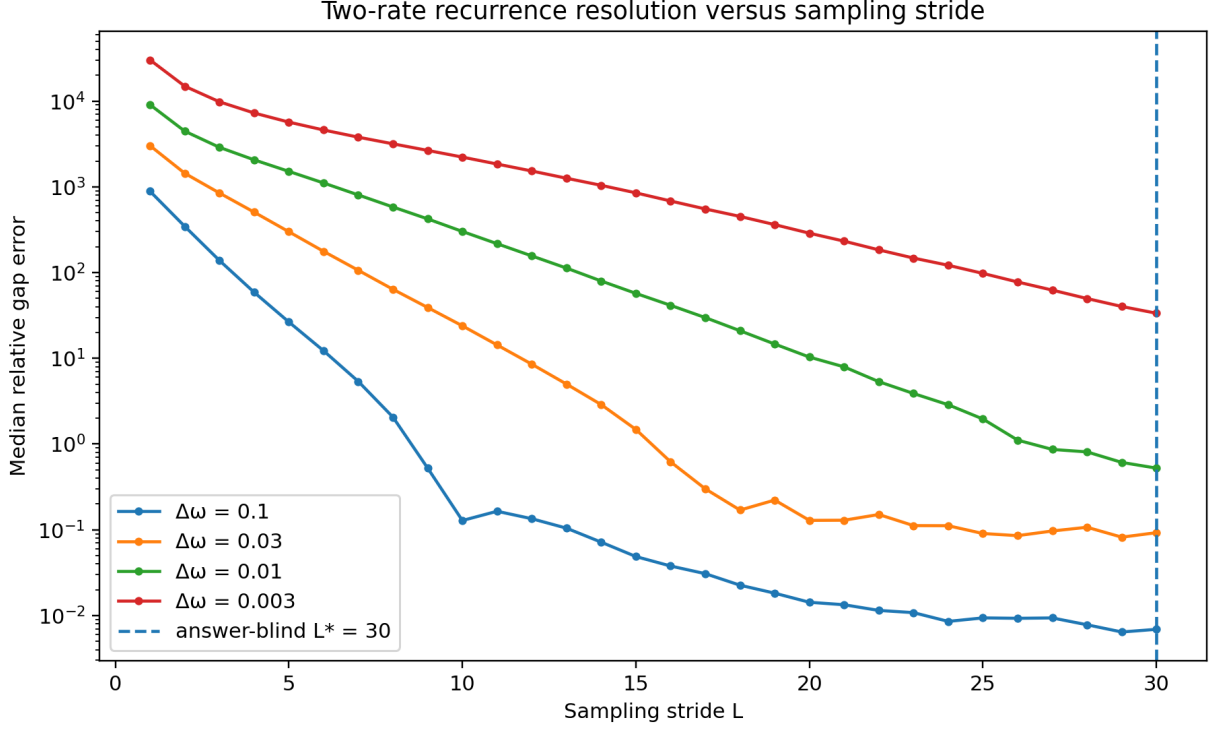


FIG. 1. Median relative gap-recovery error versus sampling stride. The vertical marker is the answer-blind stride selected from the exact discriminant rather than from observed recovery error.

The numerical frontier is not universal: it depends on duration, cadence, noise, amplitudes, estimator, and alias constraints. What survives those choices is the exact recurrence and its lag-dependent discriminant.

TABLE I. Deterministic-seed noisy benchmark at the answer-blind selected stride. Errors are relative to the true frequency gap; 120 trials were used per row.

$\Delta\omega$	median relative error	95th percentile
0.10	0.00690	0.02448
0.03	0.09252	0.23562
0.01	0.52350	1.43447
0.003	33.54698	48.21774